

AlgoChowk — Research Note

NIFTY 50 Event Study: Recovery After Significant One-Day Falls

Hypothesis. After a significant one-day fall in NIFTY 50, the index may show positive recovery over the following few trading days. The investigation is designed to measure this relationship rather than assume that it is tradable.

Event definition. A qualifying event is a NIFTY 50 close-to-close return less than or equal to -2%. The -2% threshold is a simple, reproducible starting definition of a large daily decline. The notebook keeps the threshold configurable for robustness checks.

Entry / exit. The event is identified only after the event-day close. Therefore, the default entry is the next trading day's Open, which avoids look-ahead bias. Forward returns are calculated for 3, 5 and 10 trading days. The simple event-driven backtest uses a 5-day holding period.

Data and validation. The working notebook uses daily NIFTY 50 OHLC data and checks chronological ordering, duplicate dates, missing OHLC values, positive price values and actual date coverage. The data source and downloaded date range should be retained/documented with the submitted repository.

Costs. A configurable 5 basis-point round-trip transaction-cost assumption is included. Slippage and the effect of different cost assumptions are limitations to discuss rather than reasons to optimise the strategy.

Statistics. For each holding period the notebook calculates number of observations, mean, median, win rate and standard deviation. These describe both the typical outcome and the variability of outcomes. A one-sample t-test is included as an initial statistical check; its p-value should not be interpreted as proof of economic significance.

Baseline and robustness. Event-period returns are compared with normal NIFTY forward returns over the same sample. The event threshold is varied across -1.5%, -2%, -2.5% and -3% while keeping the research engine unchanged. Holding periods are also configurable.

Out-of-sample validation. The notebook splits the historical sample chronologically: the first 70% is development/research data and the final 30% is held out as out-of-sample data. The out-of-sample event count and return statistics are reported separately.

Backtest. A deliberately simple event-driven backtest enters at the next day's Open and exits after five trading days. Overlapping signals are skipped so that closely occurring events do not create simultaneous positions in the simple test. The notebook reports number of trades, cumulative performance and maximum drawdown.

Key assumptions and risks. Results can be affected by sample size, market regimes, overlapping events, transaction costs, slippage, data quality, look-ahead bias, data snooping and post-hoc parameter selection. A positive historical result would not by itself establish a reliable trading edge.

Conclusion. The conclusion should be based on the generated event statistics, baseline comparison and out-of-sample evidence. The assignment explicitly permits a negative result. The appropriate conclusion is therefore empirical: whether the observed recovery is sufficiently consistent and robust to deserve further investigation.

Reproducibility. Run the supplied Jupyter notebook from top to bottom. Threshold, holding periods and transaction costs are exposed as parameters so the experiment can be repeated without rewriting the research engine.